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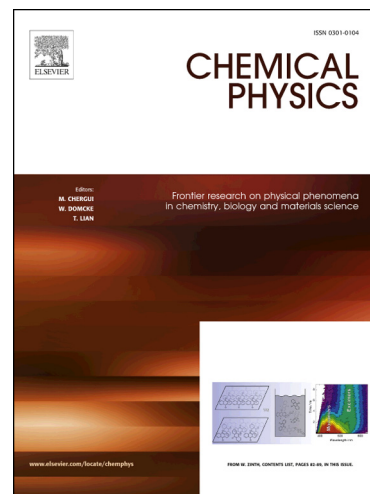
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Dielectric Properties of Acetonitrile Confined in Carbon Nanotubes

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Abstract

This study addresses numerical investigation of dielectric properties of acetonitrile confined in Single Walled Carbon Nano Tubes (SWCNTs). Frequency dependent permittivity have been computed by assuming whether interactions among the Acetonitrile (ACN) molecules exist or not. Both axial and radial dielectric permittivity show tube diameter dependence. Static dielectric permittivity becomes larger with the diameter of the tubes but still smaller than that of the bulk state as well. Furthermore, an anomalous decrease in radial component of dielectric permittivity was observed. Bending modes and stretching modes between the Carbon atoms affect the variation of dielectric permittivities as well. Moreover, Kirkwood correlation factor, g factor, firmly influenced by the size of the tube and Knudsen region where immediate interactions of ACN molecules start to develop in the this region for narrower tubes too. Both relaxation and reorientation of Acetonitrile inside the CNTs are slower than those of the bulk state.

Keywords: Acetonitrile, SWCNT, Molecular Dynamics, Dielectric

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1. Introduction

Despite both theoretical and experimental studies about dielectric phenomenon of neat Acetonitrile and its mixtures are available for a long time, the microscopic point of view about its behavior under confinement is not well understood if one considers its usage in electrolyte solutions in favor of lower viscosity. Uniform translational diffusion, relatively high dielectric permittivity and conductivity of the acetonitrile are other advantages for capacitors applications [1, 2, 3]. The use of less viscous organic solvents with ionic liquids promotes penetration in nan pores, which greatly enhances the operating voltage, the energy density and higher rate scalability of the capacitor [4, 5, 6, 7, 8]. Lithium perchlorate in acetonitrile enabling obstruction free nano-pores [9], hybrid capacitors operating high temperatures with high cycle performance [10, 11] and nano porous carbons in an electrolyte containing acetonitrile showing good performance under extreme conditions [12] are some applications pointing out importance of acetonitrile in such a fast developing technology of supercapacitors.

Although there are wide range of interests for Acetonitrile, studies investigating its dielectric behavior in molecular scale is rather limited. Recently, in reference [13], short time behavior of Acetonitrile is determined by libration of molecules and, with a larger time, the molecular motion is influenced by rotational diffusion. Investigating the capability and accuracy of some proposed models compared with the experiments [14, 15, 16, 17], structure organization of ACN and ionic liquid mixtures in nanoscale [18], dielectric relaxation of mixtures including ACN [19, 20, 21] are some studies appeared in last two decades.

Although we could not address all studies concerning with the role of ACN in electrolyte mixtures and its interaction with nano-structured electrodes, use ACN in reducing the viscosity of electrolytes seems to be significant. To the author's knowledge, there is no study addressing dielectric relaxation phenomena in confinement. These are the reasons why this study aims to investigate dielectric properties and structural organization ACN in SWCNTs. The second part of our study is organized as follows. After briefly mentioning computational

methodology in the next section, dielectric properties are given each separate section followed by a conclusion.

2. Modeling and computational methodology

Acetonitrile was represented by a six-site model with three methyl hydrogen
 35 atoms (H), a methyl carbon atom (C1), a nitrile carbon atom (C2), and a nitro-
 gen atom (N). Forces acting on each atom are sum of intra and intermolecular
 one, $E_{val} + E_{nb}$, where E_{val} and E_{nb} express intramolecular and intermolecular
 contributions respectively. Pairwise interactions accounted for dispersive and
 electrostatic interactions are written as

$$E_{nb}(r_{i\alpha j\beta}) = 4\epsilon_{i\alpha j\beta} \left[\left(\frac{\sigma_{i\alpha j\beta}}{r_{i\alpha j\beta}} \right)^{12} - \left(\frac{\sigma_{i\alpha j\beta}}{r_{i\alpha j\beta}} \right)^6 \right] + \frac{Cq_{i\alpha}q_{j\beta}}{\epsilon_0 r_{i\alpha j\beta}}. \quad (1)$$

40

Here, $r_{i\alpha j\beta}$ represents the distance between the atom i in molecule α and the
 atom j in molecule β . σ and ϵ are the equilibrium atomic separation and the
 energy well respectively. q and C stand for partial charge of an atom and
 Coulomb's constant respectively. Electrostatic and dispersive interactions have
 45 also been accounted in 1-4 interactions within an ACN molecule. Lorentz-
 Berthelot mixing rule was applied for determining cross interactions between the
 atoms. Not that all force field parameters were adopted from DREIDING/A [22]
 and summarized in Table 1. Dipole moment equals to 3.88 D based on the
 parameters given in Table 1.

50 It should be noted that bond stretching, angle bending and torsion interac-
 tions were considered in modeling intramolecular energy and force evaluations
 of CNTs. Intermolecular potential parameters of C atoms of the CNT were
 treated as of C1 atom of the ACN for evaluation of CNT-ACN interactions.

All simulations were carried on the large scale atomic/molecular massively
 55 parallel simulator (LAMMPS [24]).

The simulations were initialized with the minimization step employing Con-
 jugate Gradient method (at least 10000 number of iterations has been allowed)

Table 1: Force field parameters for acetonitrile

Harmonic bond stretch parameters	r_{eq} (Å)	k_{ij} ((kcal/mol)/Å ²)	
H-C1	1.090	665.0	
C1-C2	1.450	700.0	
C2-N	1.192	2500.0	
C-C (CNT)	1.420	700.0	
Harmonic angle bend parameters	θ_{eq} (deg)	k_{ijk} ((kcal/mol)/rad ²)	
H-C1-H	109.471	100.0	
H-C1-C2	109.471	100.0	
C1-C2-N	180.000	55.0	
C-C-C (CNT)	120.000	100.0	
Torsion for CNT	φ_{jk}^0 (deg)	V_{jk} (kcal/mol)	n_{jk}
	0	22.50	2
Lennard-Jones parameters and partial charges based on MP2/6-311+G* [23]			
	ϵ (kcal/mol)	σ (Å)	q
C1	0.0950	3.457	-0.552
C2	0.0950	3.457	0.475
H	0.0152	2.2147	0.190
N	0.1490	3.292	-0.493

with convergence criteria of 1.0×10^{-8} and 1.0×10^{-10} for energy and force respectively. Long range electrostatic interactions were calculated by the Particle-Particle, Particle Mesh (PPPM) method [25] with a cut-off of $13\text{\AA}$ and a relative error, 1.0×10^{-5} , in force evaluations. The LJ cut-off used in simulations was also $13\text{\AA}$. The dielectric constant has been set to unity and neighbor list has been updated in every 10 time steps.

Simulations were performed in two stages. In the first stage of the simulations, armchairs Single-Walled Carbon Nano Tubes (SWCNTs) with five different diameters were filled with ACN. In the second stage, systems of ACN filled SWCNTs were simulated. Equations of motion in LAMPPS are solved by time-reversible measure-preserving Verlet or rRESPA integrators depending on the ensemble (NVT or NPT) used. According to type of mentioned stages the following computational strategies were employed. The following section describes how the carbon nanotubes were filled with acetonitrile.

2.1. Filling Process

Open-ended CNTs whose length is $49.19\text{\AA}$ were immersed in a simulation box in size $50.88 \times 50.88 \times 101.77\text{\AA}^3$ which is full of ACN molecules at bulk density, 777 kg/m^3 , temperature, 298 K . Overlapping molecules of ACN with the atoms of CNT were shifted properly by our home-made code so that bad contacts are eliminated. NPT ensemble has been employed during the filling process, and the integration time step was 0.5 fs . The system was heated to 298 K starting from 10 K by applying the Noose-Hoover thermostat with 25 fs coupling time. Nose-Hoover barostat with 250 fs coupling time was used to set the pressure 1 atm . After 60 ps period of warming and pressure regulation time, the system was well equilibrated over 300 ps period of time while center of mass location of the CNT was fixed spatially in each 25 fs . Once filling process taken place, molecules whose center of mass locations are outside of the CNT were excluded. Then, the CNT and ACN molecules were doubled by imitating the original system. In this case, the length of the CNTs was increased to $98.38\text{\AA}$.

Table 2: Geometries of systems and densities of confined ACN

System	System dimensions (Å)	Number of ACN molecules in the system	Density of ACN (kg/m ³)
Bulk ACN	$50.88 \times 50.88 \times 50.88$	1500	777.0
ACN + CNT (10,10)	$d = 13.55, l = 98.38$	96	460.61
ACN + CNT (15,15)	$d = 20.33, l = 98.38$	270	575.77
ACN + CNT (20,20)	$d = 27.11, l = 98.38$	534	640.54
ACN + CNT (30,30)	$d = 40.67, l = 98.38$	1274	679.18

Details of resulting ACN-CNT systems and densities¹ were tabulated in 2. In the proceeding section we describe the method of data production simulations for analyzing the interactions between the molecules of acetonitrile and carbon
 90 nanotubes.

2.2. Simulations of ACN inside CNT

CNTs were located on a rectangular parallelepiped box where periodic boundary conditions were imposed in each direction. The longitudinal length of the box was set to 98.38 Å so that interactions of ACN molecules with surround
 95 are taken place just on each open ends of CNTs. Simulation procedure for ACN-CNT couple is the same as that of filling process with some exceptions. Systems were well equilibrated over 200 ps after the temperature was set to 298 K within 100 ps, and then data were collected in every 5 fs and over 500 ps for both velocities and positions of ACN molecules. Molecular strain rate tensors
 100 were recorded in every 0.5 fs over 1.5 ns while center of mass velocity of CNT was removed in every 150 fs.

In the proceeding sections, we would like to introduce dielectric properties of ACN systematically.

¹ACN molecules were assumed to occupy the confinement volume drawn by CNTs

3. Dielectric relaxation

105 In this section, the authors will discuss dielectric phenomenon of the ACN under confinement. For this purpose, the system is assumed to be a isotropic, polarizable, homogeneous and continuous media. Furthermore, an external and frequency dependent electric field, $\mathbf{E}(\nu)$, is acting on it.

Now, electric displacement vector can be written as the sum of externally acting electric field and polarization field - i.e.,

$$\mathbf{D}(\nu) = \epsilon_0 \mathbf{E}(\nu) + \mathbf{P}(\nu), \quad (2)$$

where ϵ_0 stands for dielectric constant of vacuum. If the dielectric properties of the system are assumed to be linear and isotropic, the polarization, $\mathbf{P}$, is proportional to the electric field, $\mathbf{E}$, and given by

$$\mathbf{P}(\nu) = \epsilon_0 \mathcal{X}(\nu) \mathbf{E}(\nu) \quad (3)$$

where $\mathcal{X}(\nu)$ represents frequency dependent electric susceptibility. Substituting Eqn 3 into Eqn 2 leads to

$$\mathbf{D}(\nu) = \epsilon_0(1 + \mathcal{X}(\nu)) \mathbf{E}(\nu) = \epsilon_0 \epsilon \mathbf{E}(\nu) = \epsilon_a(\nu) \mathbf{E}(\nu), \quad (4)$$

$$\epsilon(\nu) = \frac{\epsilon_a(\nu)}{\epsilon_0} \quad (5)$$

where $\epsilon_a(\nu)$ and $\epsilon(\nu)$ are called complex absolute permittivity and complex relative permittivity of the system respectively. The complex relative permittivity is written simply

$$\epsilon(\nu) = 1 + \mathcal{X}(\nu) \quad (6)$$

whose real and imaginary parts, $\epsilon(\nu) = \epsilon'(\nu) - i\epsilon''(\nu)$, correspond to the relative permittivity and the dielectric loss of the system respectively.

110 In this study, complex permittivity has been evaluated for two limiting cases, either for collective dynamics among the ACN molecules or for vanishing interactions among them (see discussions by McQuaarrie at Ref. [26], on page 474 and pages 495-497 for further discussions).

Due to confinement of the ACN, one can expect that the dielectric permittivity depends on the orientation inside the tube. Therefore, effective dielectric permittivity can be defined as discussed in Ref. [27],

$$\epsilon(\nu) = \frac{2\epsilon_r(\nu) + \epsilon_z(\nu)}{3} \quad (7)$$

115 where ϵ_r and ϵ_z represent radial and axial dielectric permittivity respectively.

Dense system approach: In the first case, when one considers the system as a general case that a system is dense and each dipole moment is correlated with each other, then the system is assumed to exhibit a collective dynamics in which there is a dipole-dipole interaction among the molecules. With application of linear response theory for equilibrium, the electric susceptibility in 3D is given by

$$\chi(\nu) = \frac{1}{k_B T} \frac{\langle \mathbf{M}(0) \cdot \mathbf{M}(0) \rangle}{V \epsilon_0} \mathcal{L}_{iw}\{-\dot{\Phi}\}, \quad (8)$$

where the operator $\mathcal{L}_{iw}$ stands for Fourier-Laplace transform, i.e.

$$\mathcal{L}_{iw}\{-\dot{\Phi}\} = \int_0^\infty e^{-i\nu t} \left(-\frac{d\Phi(t)}{dt} \right) dt \quad (9)$$

and $\mathbf{M}(t)$ is vectorial sum of dipole moments of constituent molecules at time t . k_B and T stand for Boltzmann constant and thermodynamic temperature respectively. V is the volume enclosed by CNTs. In this case, the correlation function is treated as

$$\Phi(t) = \frac{\langle \mathbf{M}(0) \cdot \mathbf{M}(t) \rangle}{\langle \mathbf{M}(0) \cdot \mathbf{M}(0) \rangle}. \quad (10)$$

120 In Eq. 10, the system is assumed to be a dense system where all interactions among the molecules have been taken account. Thus, Eq. 8 leads to a solution, complex permittivity, that interaction of the dipole moment of each molecule is correlated with other molecules.

Dilute system approach: In the second case, cross terms between the dipole moments on different molecules are vanished and the system is assumed

to be a dilute system in which the dynamics of each dipole moment is independent from the others. In such case, the correlation function is expressed in terms of orientation of the dipole moment of each molecule such that

$$\Phi(t) = \frac{\left\langle \sum_{i=1}^{N_m} \mathbf{u}_i(0) \cdot \mathbf{u}_i(t) \right\rangle}{\left\langle \sum_{i=1}^{N_m} \mathbf{u}_i(0) \cdot \mathbf{u}_i(0) \right\rangle}, \quad (11)$$

followed by the electric susceptibility

$$\chi(\nu) = \frac{1}{k_B T} \frac{\left\langle \sum_{i=1}^{N_m} \mathbf{u}_i(0) \cdot \mathbf{u}_i(0) \right\rangle}{V \epsilon_0} \mathcal{L}_{i\nu}\{-\dot{\Phi}\}, \quad (12)$$

where N_m and $\mathbf{u}$ are the number of the molecules in the system and the dipole moment of each molecule respectively.

In the following section, the authors show how the interaction mechanism among the individual dipole moments affects the dielectric behavior in the bulk and confined ACN.

4. Results and Discussions

Laplace-Fourier transforms appeared in both Eqn 8 and 12, depending on the assumption of interaction mechanism, are computed on the raw data spanning 0 to 50 ps to evaluate the relaxation phenomenon inside the nano tubes. The time step $\Delta t = 5$ fs. The effect of tube diameter on both $\epsilon_r(\nu)$ and $\epsilon_z(\nu)$ is illustrated in Figures 1-4. Moreover we illustrated the effect of mode frequencies on the variation of dielectric permittivity of ACN inside CNT (30-30) in Figure 5.

Analyses show that the interacting molecules approximation leads to reliable estimate of axial the dielectric constant confined in CNT (30,30) compared with experimental datum, 36.55, as of Ref. [28] (see Table 6) while the dielectric constant based on the dilute system approach doesn't show mutual consistency with the experimental one. It is obvious that the density is a managing physical

parameter in the magnitude of the axial dielectric constant. Especially for wider tubes, the liquid shows bulk behavior in the axial direction. The dielectric constant computed for ACN inside CNT (30-30), 37.10, is close the experimental bulk value, 36.55. Contrary to observation made by Lin and co-workers reported in [27], the tubes with small diameters possess lower axial permittivity. We think that as the density becomes smaller, the molecules have more space to move freely. The dipoles are not forced to align themselves in the axial direction. In radial direction, dielectric permittivity were all dropped drastically to almost vacuum permittivity. Such phenomenon was observed by some authors. For example, Chaban has attributed small values of dielectric constant in CNTs under consideration to single file pattern due to confinement relatively narrow tubes. Orientation of dipoles is crucial in determining radial dielectric permittivity. Figure 7 shows how g-factor varies along the radial direction of CNTs. In radial direction, no preferred orientation is observed at far away from the wall of the CNTs. However, such orientations are almost close to each other. This behavior may lead to drastic decrease in radial dielectric permittivity.

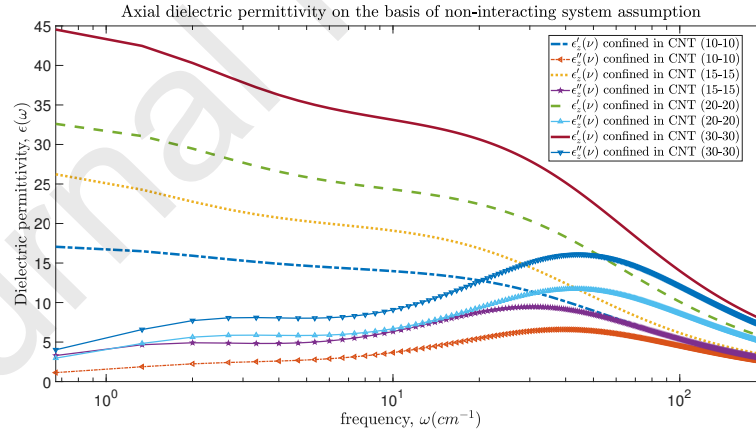


Figure 1: Frequency dependent dielectric permittivity in axial direction based on dilute system approaches.

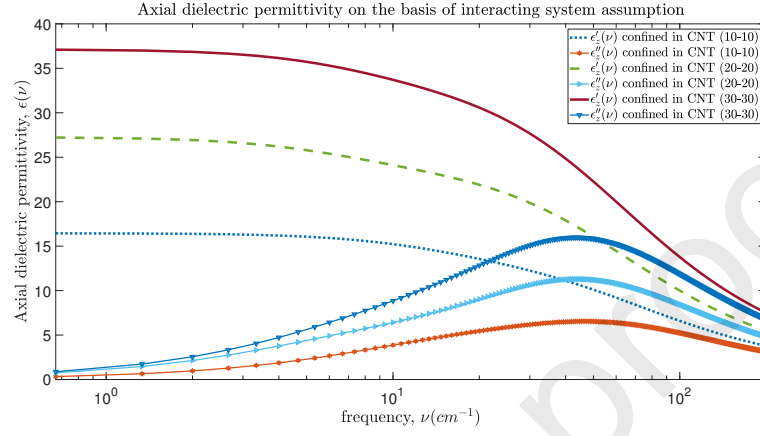


Figure 2: Frequency dependent dielectric permittivity in axial direction based on dense system approaches.

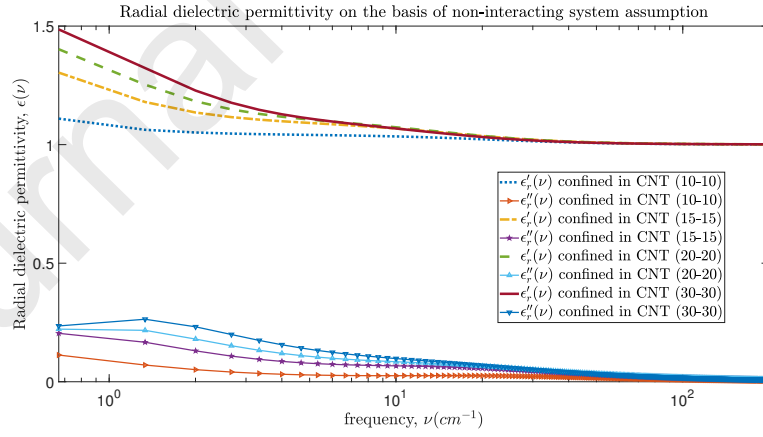


Figure 3: Frequency dependent dielectric permittivity in axial direction based on dilute system approaches.

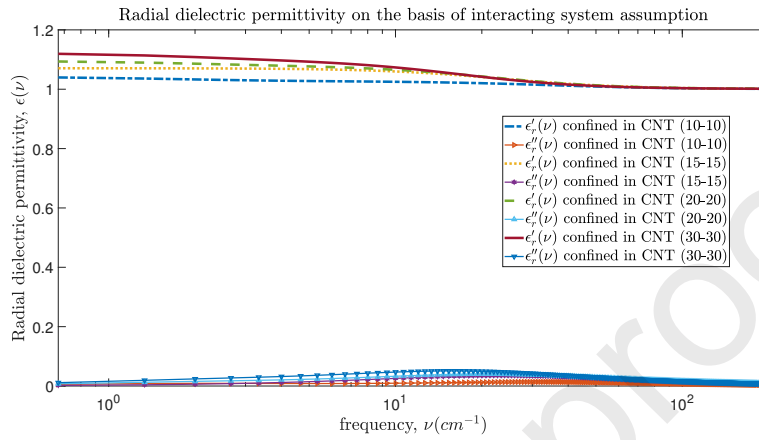


Figure 4: Frequency dependent dielectric permittivity in axial direction based on dense system approaches.

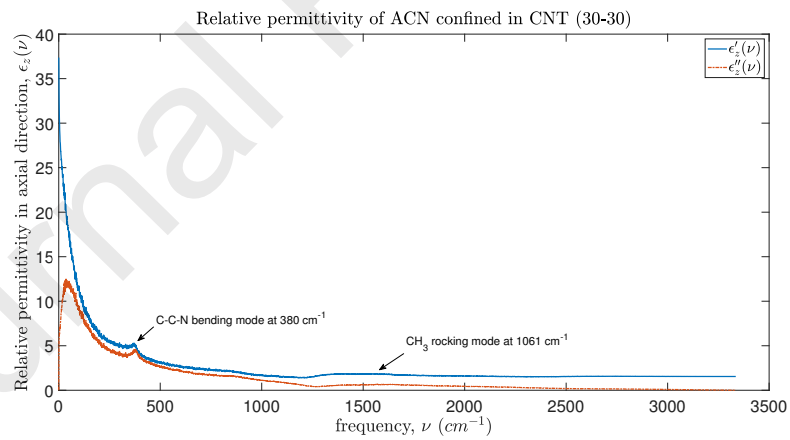


Figure 5: Dielectric constant and loss of ACN confined in CNT(30-30) as a function of frequency. In this graph the system is assumed to be composed of interacting molecules. Notice that, the effect of C-C-N bending mode at 380 cm^{-1} and CH_3 rocking mode at 1061 cm^{-1} on the variation of dielectric constant and loss are easily identified.

Table 3: Computed static dielectric constants of the Acetonitrile at $T = 298$ K. Laplace-Fourier transforms performed on data with $\Delta t = 5$ fs and number of time steps was 10000. Here, *dense* and *dilute* stand for computations based on the assumptions of interacting molecules and non-interacting molecules respectively

Dielectric constant	CNT(10,10)	CNT(15,15)	CNT(20,20)	CNT(30,30)
$\epsilon'_{r,dense}(0)$	1.0416	1.07	1.095	1.121
$\epsilon'_{z,dense}(0)$	16.400	21.014	27.22	37.100
$\epsilon'_{dense}(0)$	6.161	7.718	9.803	13.11
$\epsilon'_{r,dilute}(0)$	1.301	1.287	1.549	1.608
$\epsilon'_{z,dilute}(0)$	17.310	27.320	33.290	44.540
$\epsilon'_{dilute}(0)$	6.637	9.964	12.129	15.918

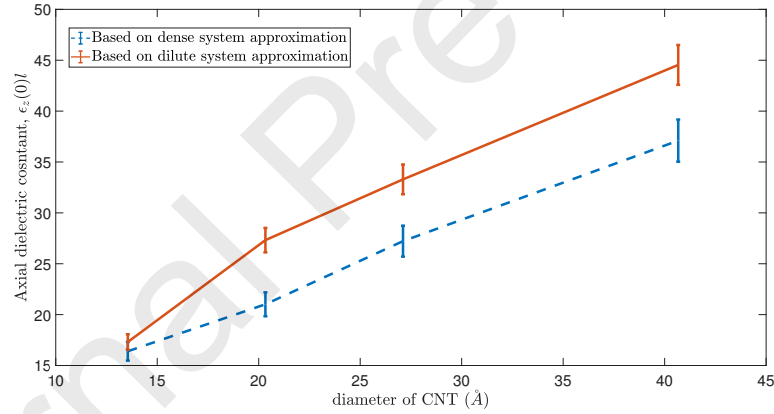


Figure 6: A clear demonstration of how static dielectric constant in axial direction varies with CNT's diameter.

At this stage, we would like to outline a discrepancy between the behavior of water and ACN confined in nano tubes. Although dielectric constant of the water in SWCNT tends to decrease as the diameter of the tube increases [27], it decreases considerably in a spherical nano cavity of about 12 Å in diameter as reported in reference [29]. Contrary to water in SWCNT, the dielectric constant of the ACN increases with the size of the tube as shown in Fig. 6. Moreover,

the dielectric constant predicted for all cases under consideration is still smaller
 165 than bulk one (the calculated and experimental one[30]). In other words, the
 effect of confinement is still appreciated even for large sizes of the tubes.

In order to make clear why such increase in dielectric constant was observed
 with increasing diameters of tubes, Kirkwood correlation factor[31], g factor,
 was introduced ;

$$g = \sum_{i=1}^{N_m} \langle \cos \theta_{ij} \rangle, \quad (13)$$

where θ_{ij} denotes the angle between the orientations of the i^{th} and the j^{th}
 dipole. The calculated correlation factor for increasing radii in each CNT is
 shown in Fig. 7. At first glance, it can be observed that there is no correlation

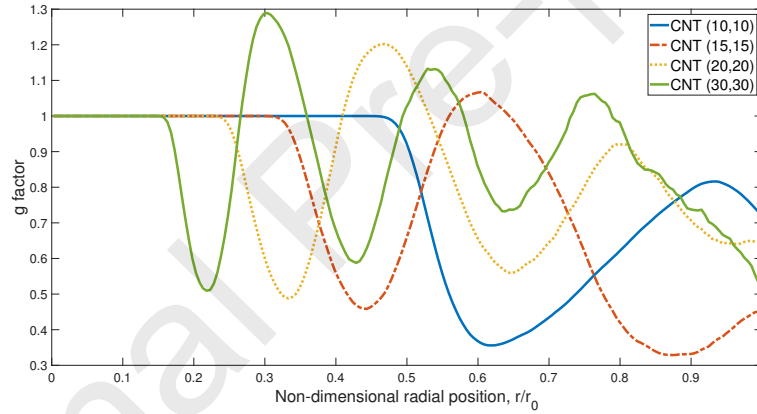


Figure 7: Radial variation of Kirkwood correlation factor calculated from the Eq. 13. Note
 that the radius of each CNT is set to unity.

170 between the i^{th} molecule and the others within the first half of the radius of the
 CNT (10-10). This is due to strong rarefaction effects observed in the core of
 this CNT. This region wherein rarefaction effects are observed becomes smaller
 with increasing sizes of CNTs. Beyond this critical radius, correlations starts to
 develop and can be represented as continuous functions. In the smallest tube,
 175 the orientation of molecules does not change much in a half of CNT's radius.
 For larger tubes, molecules orientations oscillate within a certain angle span.

This behavior slows down in the vicinity of the wall. It is clear that the coupled dynamics of both rarefaction and the confinement effect decreases the dielectric constant enormously. Although the confinement in a smaller CNT encourages
 180 a self-organized and directed dipoles in the direction free to move, in the axial direction of nano tubes, a dominant rarefaction effect tries to align rather limited number of molecules opposite to those of surface region. We believe that the main reason for considerable drops in dielectric constant is increasing fractional volume of rarefaction with smaller sizes of the CNTs.

185 So far, the results of dielectric relaxation theory and obtained frequency dependent dielectric constant for both under the assumptions of dilute and dense systems were introduced. In addition to these, the dielectric constant was evaluated as $\nu \rightarrow 0$ in Eq. 9 in conjunction with Eqn. 10, (see Ref. [32]),

$$\lim_{\nu \rightarrow 0} \int_0^{\infty} e^{-i\nu t} \left(-\frac{d\Phi(t)}{dt} \right) dt = 1. \quad (14)$$

Thus, Eqn 6 simplifies to

$$\epsilon'_z(0) = 1 + \frac{1}{k_B T} \frac{\langle M_z^2 \rangle}{V \epsilon_0} \quad (15)$$

$$\epsilon'_r(0) = 1 + \frac{1}{2k_B T} \frac{\langle M_r^2 \rangle}{V \epsilon_0}. \quad (16)$$

Here, M_z and M_r are the total dipole moment in z and radial directions of the
 190 system. The convergence of the ensemble average is known to take place in a long period of time. For this reason, in our simulations, the averages have been taken over 500 ps which seem to be sufficient by inspecting asymptotic behavior of the dielectric constant with respect to time (see the Fig. 8). Standard deviations were also too small in our calculations.

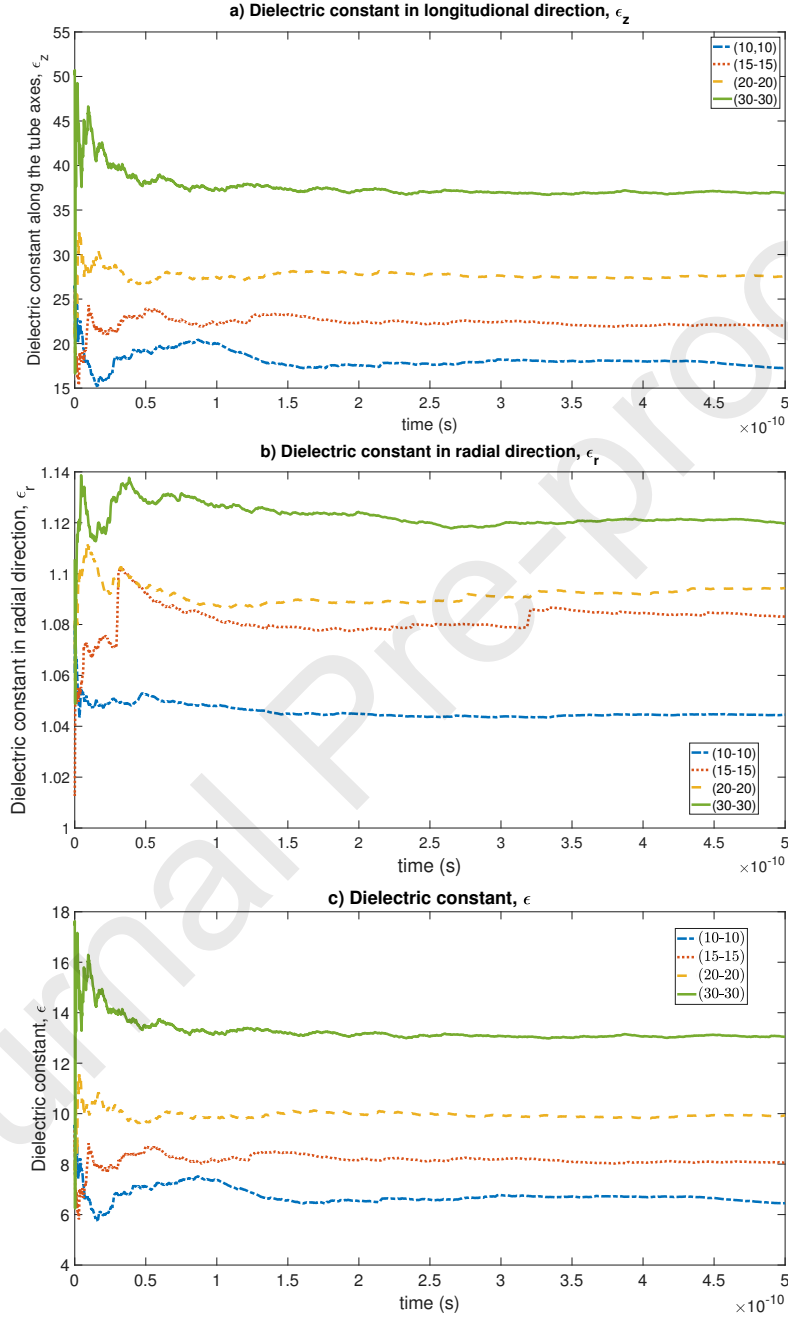


Figure 8: Dielectric constants obtained from the fluctuation formulas Eqns. 15 and 16; a) for axial direction b) for radial direction, c) for isotropic system.

Table 4: Static dielectric constants by Eqns. 15 and 16.

The system	Density of ACN (kg/m ³)	ϵ_r	ϵ_z	ϵ
CNT(10,10)+ACN	460.61	1.04 ± 0.08	17.80 ± 0.14	6.63 ± 0.19
CNT(15,15)+ACN	575.77	1.08 ± 0.07	22.29 ± 0.12	8.15 ± 0.15
CNT(20,20)+ACN	640.54	1.09 ± 0.05	27.63 ± 0.11	9.93 ± 0.13
CNT(30,30)+ACN	679.18	1.12 ± 0.05	37.2 ± 0.10	13.08 ± 0.11
Bulk	777.00	-	38.00	36.55 Ref. [28]

195 The static dielectric constants resulting from the limiting case were 6.63 ± 0.19 , 8.15 ± 0.15 , 9.93 ± 0.13 and 13.08 ± 0.11 for CNTs (10-10), (15-15), (20-20) and (30-30) respectively. The results obtained from the fluctuation formula are good agreement with those of relaxation theory.

200 In order to elucidate the effect of vibration spectra on the variation of dielectric permittivity, we show the power spectral density of time derivative of time correlation function calculated from Eq. 11 for two extreme cases, CNT (10-10) and (30-30) (see Fig. 9). There is a significant shift in the frequency of C1-H bending mode which leads to a strong peak on both spectral densities. C1-H stretching mode also shifts on the shoulder wherein a plateau is formed at high frequencies. Such a distinction was not observed in the vibration spectra shown in Fig. 10. In frequencies smaller than 600 cm^{-1} distinctions between the modes of both cases are not distinguishable.

205 The authors question whether the dielectric relaxation of the ACN obeys Debye theory or not. The dipole moment correlation function measures the correlation between the $\mathbf{M}(t)$ with $\mathbf{M}(0)$. According to Debye, in summary, initial correlation $\langle M^2(0) \rangle$ tends to decrease exponentially to zero due to molecular interactions. Nevertheless, decay of the normalized dipole time correlation functions calculated from Eq. 10 is not of simple Debye relaxation form. Therefore the normalized dipole time correlation functions were fitted to biexponential

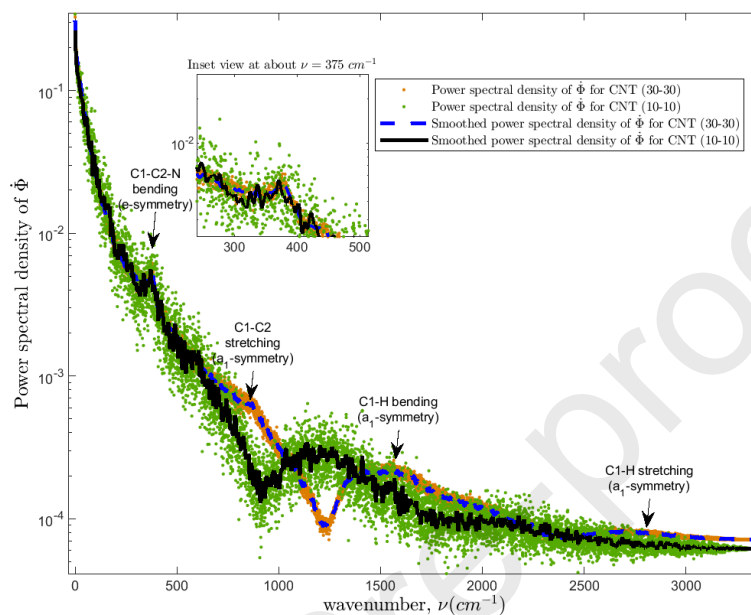


Figure 9: Power spectrum of of time derivative of time correlation function calculated from Eq. 11.

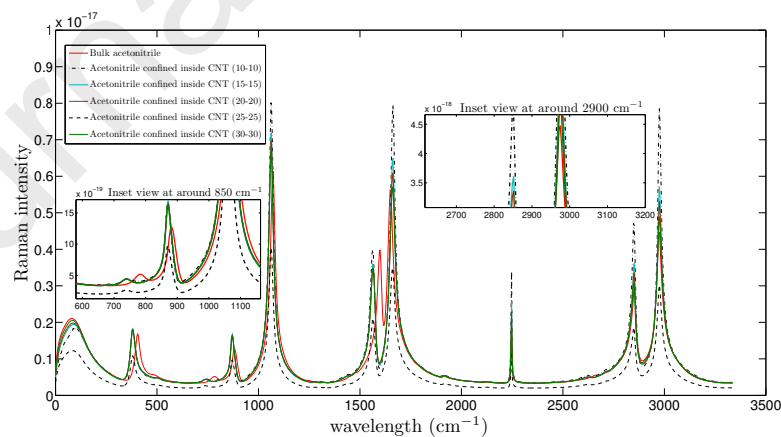


Figure 10: Variation of vibrational frequencies depending on the tube diameter.

Table 5: The Relaxation time associated with dispersion and relaxation times for the liquid acetonitrile. Biexponential function was utilized for the fitting the normalized dipole correlation functions.

	A	τ_m (s)	B	τ_n (s)	The relaxation time (s); $\langle\tau\rangle = \frac{A\tau_m^2 + B\tau_n^2}{A\tau_m + B\tau_n}$
CNT (10-10), ($R^2 = 0.968$)	0.722	2.848×10^{-13}	0.278	5.465×10^{-12}	4.847×10^{-12}
CNT (15-15), ($R^2 = 0.968$)	0.848	2.535×10^{-13}	0.152	4.69×10^{-12}	3.659×10^{-12}
CNT (20-20), ($R^2 = 0.984$)	0.787	2.290×10^{-13}	0.213	2.135×10^{-12}	1.595×10^{-12}
CNT (30-30), ($R^2 = 0.986$)	0.698	2.645×10^{-13}	0.302	1.587×10^{-12}	1.221×10^{-12}
Bulk (calculated)	0.710	1.077×10^{-13}	0.290	2.110×10^{-12}	1.900×10^{-12}

functions that are given by

$$\frac{\langle \mathbf{M}(0) \cdot \mathbf{M}(t) \rangle}{\langle \mathbf{M}(0) \cdot \mathbf{M}(0) \rangle} = Ae^{(-t/\tau_m)} + Be^{(-t/\tau_n)} \quad (17)$$

where A , B and τ_m , τ_n are dispersion terms and time parameters to be determined from fitting procedure respectively. The calculated relaxation times were summarized in Table 5.

From Table 5, it can be realized that the computed dipole time correlation function data noticeably deviate from the bulk Debye relaxation behavior. Although the faster component of dielectric relaxation τ_m is smaller than τ_n , the coefficient of the faster component, the coefficient A , seems to manage relaxation phenomena. The calculated time evolution of the dipole correlation function showed that the diffusive and slower response is missed in the initial phase of the time evolution. Although the mean relaxation time is an order of magnitude greater than the inertial component, τ_m , the dynamics of the system is dominated by the rotational diffusion of molecules and inertial effects which are taken place in a short time. In addition, there is no mutual consistency between the calculated relaxation time for bulk system which is already 1.9 ps and the experimental value, 3.35 ps, as reported in Ref. [30].

In the second way, Debye theory of dielectric suggest that $\langle \mathbf{M}(t) \cdot \mathbf{M}(0) \rangle$

decays from its initial value exponentially such that

$$\frac{\langle \mathbf{M}(t) \cdot \mathbf{M}(0) \rangle}{\langle \mathbf{M}(0) \cdot \mathbf{M}(0) \rangle} = e^{-|t|/\tau_D} \quad (18)$$

where τ_D represents Debye relaxation time. Substitution of Eqn. 18 into Eqn. 9 with accompanying Eqns. 8 and 6, gives dependency of dielectric permittivity on frequency. Thus Debye relaxation time can be stated in terms dielectric permittivity, i.e.

$$\frac{\epsilon'(\nu) - 1}{\epsilon'(0) - 1} = \frac{1}{1 + \nu^2 \tau_D^2}. \quad (19)$$

The Debye relaxation times obtained from the analyses for CNTs (10-10) and (30-30) are almost equal to 0.927 and 0.52 ps respectively. The estimate of Debye theory given by Eqn. 18 cannot capture the short time rotational behavior of molecular dipoles as reported in Ref. [33]. This is the reason why prediction of Debye theory contradicts with the inertial time component mentioned, previously.

In addition to the relaxation time, molecular reorientation dynamics in acetonitrile was analyzed through the time correlation function

$$C_l(t) = \langle P_l(\bar{\mathbf{u}}(0) \cdot \bar{\mathbf{u}}(t)) \rangle \quad (20)$$

where P_l represents l th Legendre polynomial and $\bar{\mathbf{u}}(\mathbf{t}) = \mathbf{u}(t)/u(t)$ is the unit vector of the dipole moment. The correlation functions associated with the Legendre polynomials are

$$C_1(t) = \langle \cos \theta(t) \rangle \quad (21)$$

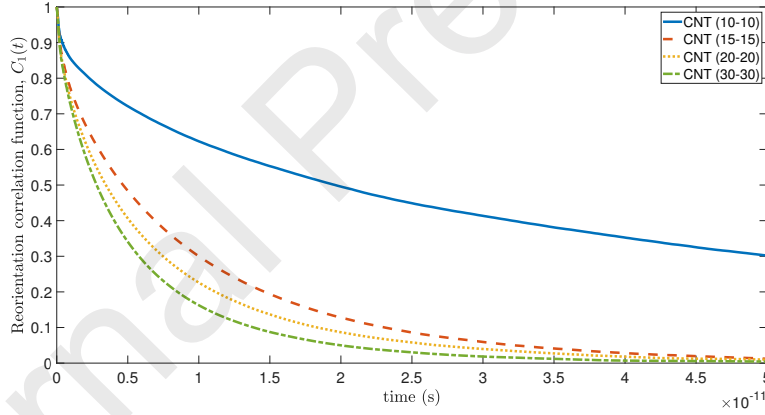
$$C_2(t) = \frac{1}{2} \langle 3 \cos^2 \theta(t) - 1 \rangle \quad (22)$$

where $\cos \theta(t) = \bar{\mathbf{u}}(t) \cdot \bar{\mathbf{u}}(0)$. Reorientation times τ_l ($l = 1, 2$) have been obtained by fitting the data produced from Eqns. 21 and 22 to triple exponential functions as of the form

$$C_l(t) = Ae^{(-t/\tau_m)} + Be^{(-t/\tau_n)} + Ce^{(-t/\tau_o)}. \quad (23)$$

Table 6: $l = 1$

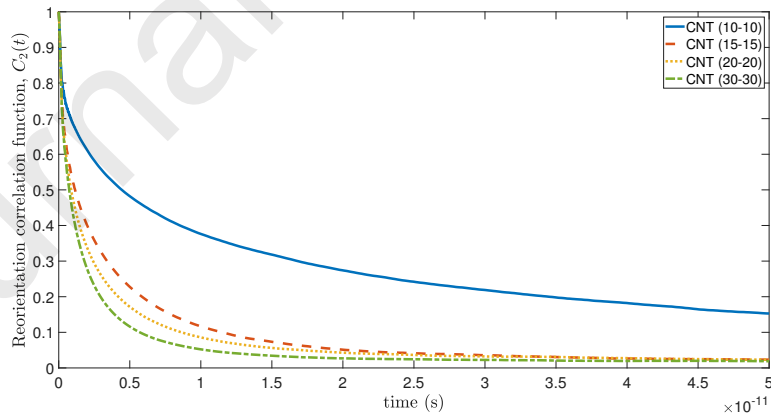
	CNT (10-10)	CNT (15-15)	CNT (20-20)	CNT (30-30)
A	0.228	0.283	0.458	0.522
τ_m (s)	5.981×10^{-12}	4.730×10^{-12}	4.290×10^{-12}	3.460×10^{-12}
B	0.678	0.562	0.388	0.355
τ_n (s)	6.082×10^{-11}	1.327×10^{-11}	1.300×10^{-11}	1.010×10^{-11}
C	-	0.149	0.150	0.125
τ_o (s)	-	3.949×10^{-13}	3.890×10^{-13}	2.700×10^{-13}
$\langle \tau_1 \rangle = \frac{A\tau_m^2 + B\tau_n^2 + C\tau_o^2}{A\tau_m + B\tau_n + C\tau_o}$ (s)	5.906×10^{-11} ($R^2 = 0.999$)	1.189×10^{-11} ($R^2 = 0.999$)	1.047×10^{-11} ($R^2 = 1.000$)	7.828×10^{-12} ($R^2 = 1.000$)

Figure 11: Reorientation correlation functions for varying tube diameters, $l = 1$.

Time evolution of $C_1(t)$ and $C_2(t)$ as shown in Figures 11 and 12 were fitted to triple exponential functions and resulting re-orientation times were tabulated in Table 6 and 7. The results show that reorientation of molecules confined in nanotubes are taken place relatively long times compared to those of bulk state ($\tau_1 = 3.28 \times 10^{-12}$ Ref. [34] and $\tau_2 = 1.02 \times 10^{-12}$ Ref. [35]). No mutual consistency for reorientation times computed for CNT (30-30) compared with

Table 7: $l = 2$

	CNT (10-10)	CNT (15-15)	CNT (20-20)	CNT (30-30)
A	0.340	0.508	0.487	0.496
τ_m (s)	5.020×10^{-12}	4.120×10^{-12}	3.320×10^{-12}	2.670×10^{-12}
B	0.398	0.0928	0.0744	0.0448
τ_n (s)	6.060×10^{-11}	3.200×10^{-11}	3.850×10^{-11}	4.770×10^{-11}
C	0.254	0.39	0.430	0.450
τ_o (s)	2.510×10^{-13}	3.400×10^{-13}	3.650×10^{-13}	3.910×10^{-13}
$\langle \tau_2 \rangle = \frac{A\tau_m^2 + B\tau_n^2 + C\tau_o^2}{A\tau_m + B\tau_n + C\tau_o}$ (s)	5.678×10^{-11}	1.996×10^{-11}	2.494×10^{-11}	2.901×10^{-11}
	$(R^2 = 0.999)$	$(R^2 = 0.999)$	$(R^2 = 0.999)$	$(R^2 = 0.999)$

Figure 12: Reorientation correlation functions for varying tube diameters, $l = 2$.

the bulk state Reorientation times decrease with increasing diameter of the tubes
 245 as well. This behavior contradicts our expectations because as the diameter of
 the tube enlarges the interactions among the ACN molecules should resemble to
 that of bulk state. Such contrasting reorientation times with the bulk state may
 be attributed to a large number of ACN molecules interacting with the CNT
 atoms on the vicinity of interface. The number of such interacting molecules
 250 depends on the diameter of the tube as expected.

So far, the authors present the effect of frequency and the tube diameter on
 the dielectric properties by different ways. Our main conclusions were given the
 proceeding section. However, we think that studying with a non-dimensional
 group of parameters presented in here may be useful for future perspectives.
 For this a purpose, one can introduce a non-dimensional parameters group that
 demonstrate competing effects between the absolute dielectric permittivity and
 viscous diffusion. The non-dimensional parameter group can be read as

$$\frac{\epsilon(0)\epsilon_0 D \nu \tau_D^2 k_B T}{u^2 d}, \quad (24)$$

where D , ν and u stand for diffusion coefficient, kinematic viscosity and the
 dipole moment of ACN for parameters given in Table 1 respectively. Kinematic
 viscosity, $\nu = \mu/\rho$ where ρ is density of ACN in confined volume, and diffu-
 sion coefficient, D are borrowed from the first part of this study (Ref. [36]).
 255 As we discussed previously absolute dielectric permittivity is influenced by the
 decreasing size of confining structure where in it partially increases with the
 diameter. Contrary to dielectric permittivity, viscosity tends to be large with
 smaller tube diameters, like diffusion coefficient. Such contending physical pa-
 rameters in an one non-dimensional group results in an important variation
 260 from its highest 2.78 to the lowest 0.021 value, depending on the tube diameter.
 This behavior elucidates how absolute dielectric permittivity is susceptible in
 defining declining trend of non-dimensional group of parameters. It should be
 clear that the viscosity and diffusion coefficient play an essential role in terms of
 its order of magnitude compared with those of the others. The variation of this
 265 non-dimensional group of parameters can give some hints to supercapacitors

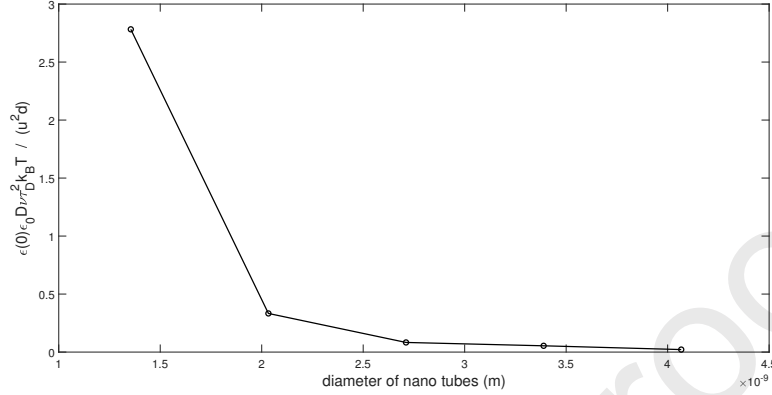


Figure 13: Variation of non-dimensional group of parameters as function of tube diameter. Diffusion coefficient, kinematic viscosity, and other parameters are evaluated at 298 K

designers.

Coating carbon nanotubes to the electrode materials is trending technique to increase specific surface area of the electrode accessible to electrolyte ions to increase capacitance. Thus, on the coated surface of the electrodes, there will be voids which resemble to interior volume of the carbon nano tubes. Due to intermolecular forces among the molecules and ions that constitute electrolyte materials, there will some volume where electrolyte molecules/ions couldn't reach. In this text, we define such volume as a Knudsen region as well. Decreasing pore size, in analogous to nanotube's interior volume, results in less electrolyte materials inside pores, and this leads to less capacitance as well. Nevertheless, there is no consensus about the optimal value of the pore size. Although, Largeot et al. [37] point out that optimal value of pore size strictly depends upon the size of ions, Simon and Gogotsi [38] report an anomalous capacitance increase in carbon electrodes where the pore size is less than 1 nm. In fact, our analyses for CNT (10-10) with a diameter 1.355 nm reveal underlying mechanism of interaction of the solvent atoms with the C atoms of the CNT. If we remember the radius of rarefied region whose value approximately an half of the radius of the CNT(10-10) (see Figure 7 for detailed view), we can eventually realize that

the distance between the atom whose potential well is the maximum among the others and C atoms of the CNT will be simply

$$\delta r = r_{pore} - \sqrt{\sigma_{max}\sigma_C}, \quad (25)$$

where σ_{max} is intermolecular equilibrium distance for an atom that has maximum potential well among the constituting atoms of the system and σ_C is the intermolecular equilibrium distance of the CNT. In our calculations δr is equal to 3.373 Å which is almost equal to an half of the radius of pore (in our case $0.50r_{CNT(10-10)} = 3.387\text{Å}$) Therefore, we propose a relation, such that

$$r_{min. \text{ of pore}} > 2\sqrt{\sigma\sigma_C} \quad (26)$$

in order to eliminate increase in viscosity, the volume of rarefied region within the pore and to ensure occupancy of solvent molecules and ions in the pore. In Eqn. 26, $r_{min. \text{ of pore}}$, σ and σ_C refer the minimum radius of the pore on the electrode surface, the Lennard-Jones equilibrium distance of the atom whose
 270 potential well is the maximum of the atoms of constituent molecules/ions and Lennard-Jones equilibrium distance of the C atom of the CNT respectively. Investigation of the pore size may be beneficial for the industry and is left to another study for more details.

275 5. Concluding remarks

This study predicts dielectric behavior of acetonitrile in SWCNTs by series of computational experiments. We have discussed how the confinement effect changes the dielectric properties of the acetonitrile at 298 K.

280 Dielectric properties of the ACN seems to be very sensitive both rarefaction and confinement effects. The magnitude of the dielectric constant is a function of the size of the dilute volume; i.e., increasing size which diminishes the dielectric constant cooperating with the confinement effect. Both radial and axial dielectric constants decreases with diameter of the tubes. No preferred orientation of the dipoles along the radial direction seems to be the other reason for low values of the radial dielectric constant. We have used two different 285 methods in the simulation of dielectric relaxation. In the first one, the system is treated as a system composed of interacting molecules. In the other one, the system is treated as individual molecules which do not affect each other. The results of both approximations are totally different. Our simulations for the 290 bulk state show that the assumption of interacting particles gives better result which constitutes well with the experimental data. Moreover, we observed that the relaxation time of the acetonitrile increases considerably compared with experiments. The relaxation process is mainly based on rotational diffusion.

Based on our further discussions, we conclude that, for better diffusion of 295 the solvent molecules/ions across the porous surface that made of CNTs, the minimum pore radius should be greater than twice the distance built between the atom whose potential well is the highest among the constituent atoms of electrolyte solution and the CNT's C atom.

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